

CS2006: 計算機組織

Course Outline



為什麼電腦不用十進位
而用二進位？

二進制只有兩種狀態

軟體、硬體

對、錯

本土化、非本土化

陰、陽

開、關

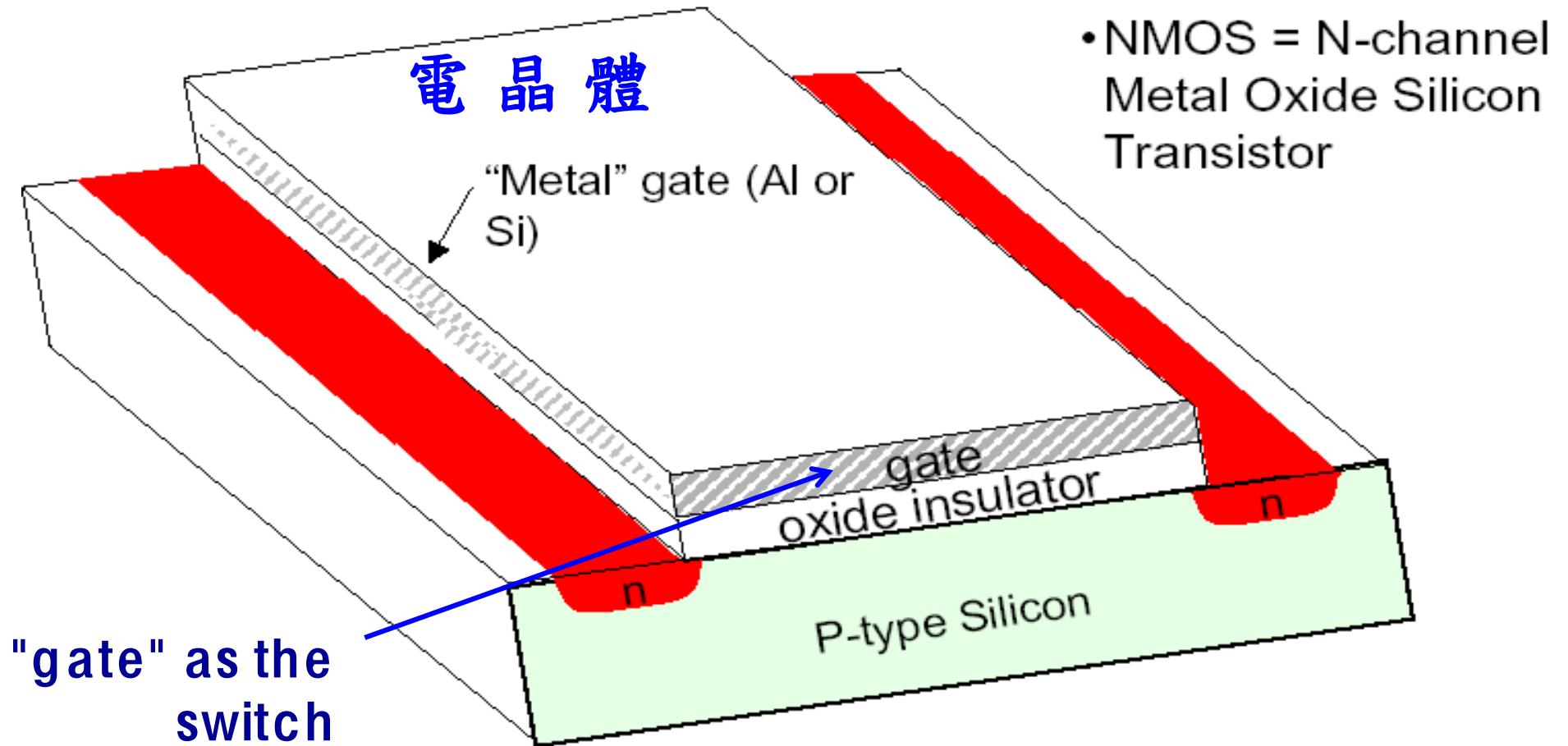
正、反

真、偽

勝、負

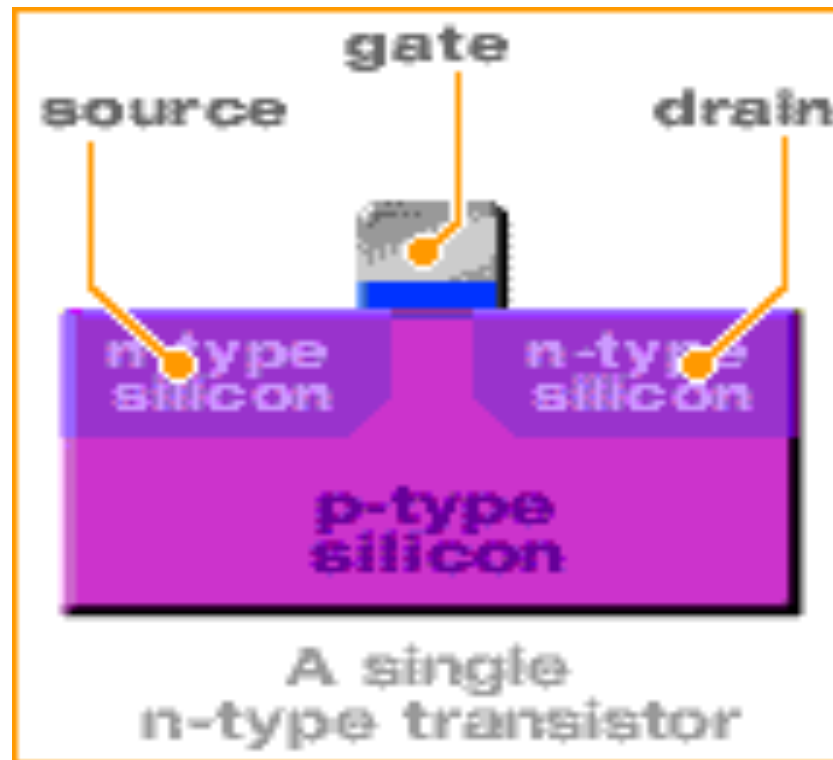
如何做出電子開關？

NMOS TRANSISTOR STRUCTURE



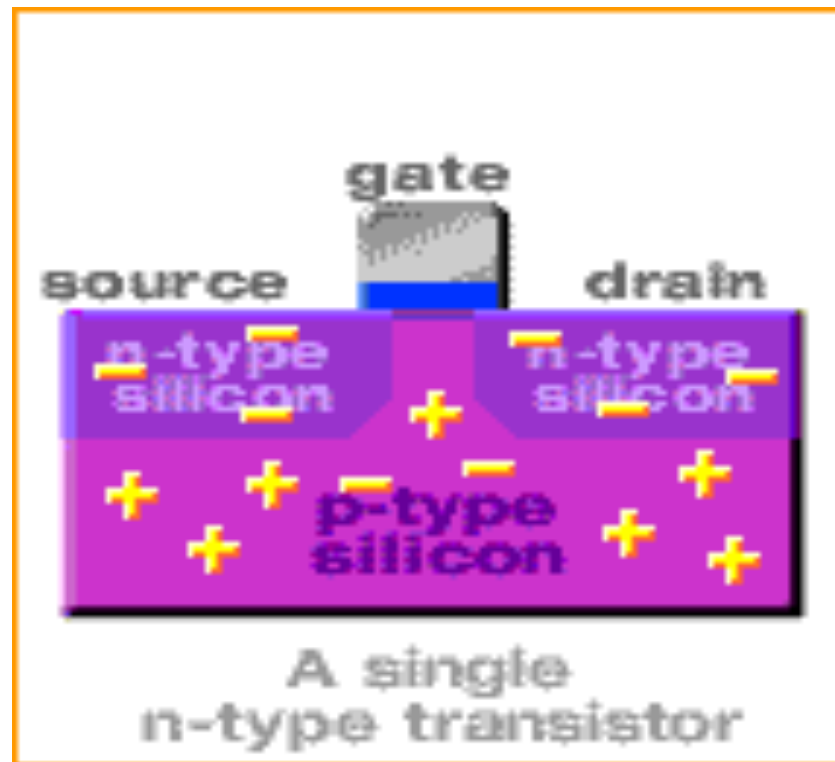
A Working Transistor (1/6)

- ◆ Transistors consist of three terminals; the source, the gate, and the drain:



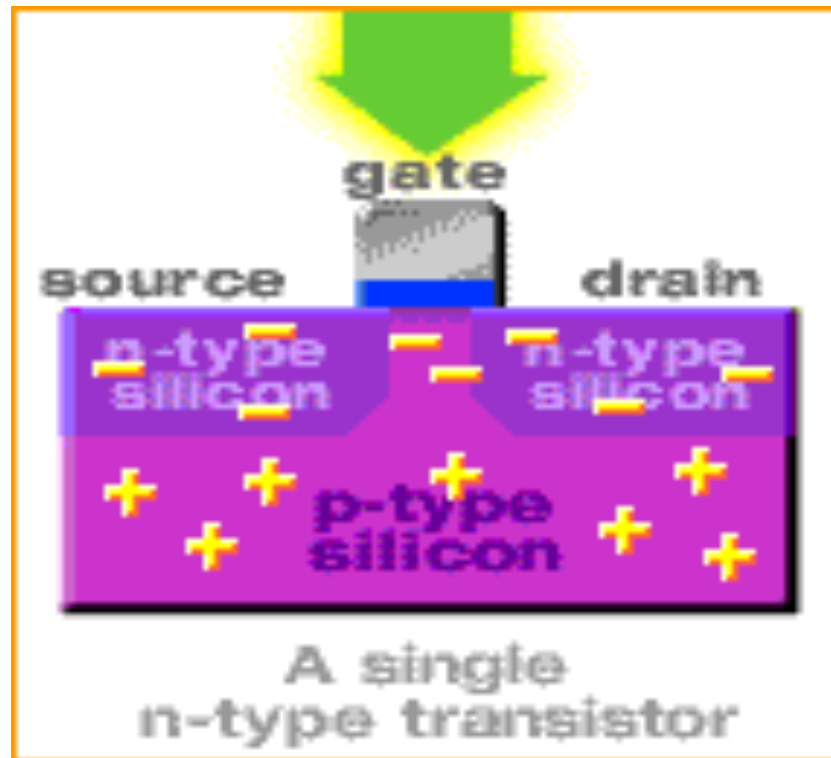
A Working Transistor (2/6)

- ◆ In the n-type transistor, both the source and the drain are negatively-charged and sit on a positively-charged well of p-silicon.



A Working Transistor (3/6)

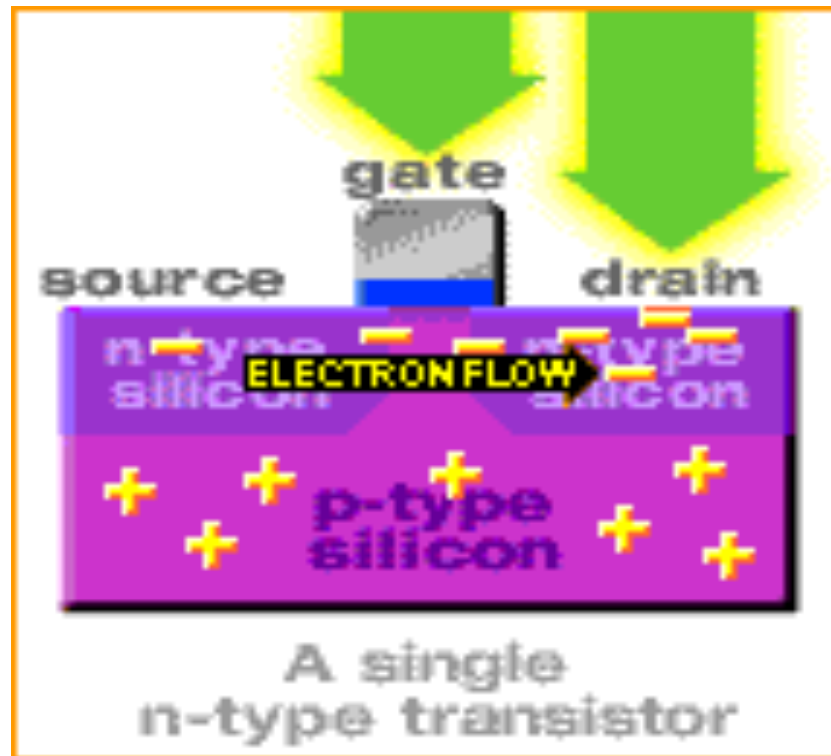
- ◆ When positive voltage is applied to the gate, electrons in the p-silicon are attracted to the area under the gate forming an electron channel between the source and the drain.



A Working Transistor (4/6)

- ◆ When positive voltage is applied to the drain, the electrons are pulled from the source to the drain. In this state the transistor is on.

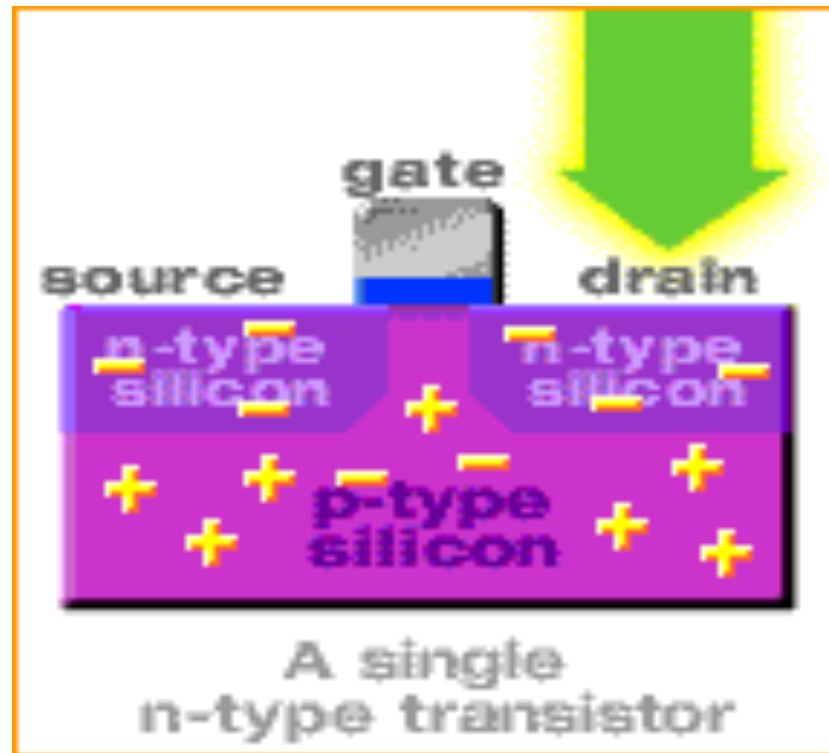
開



A Working Transistor (5/6)

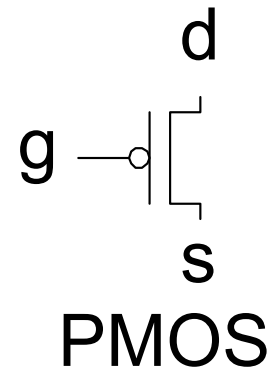
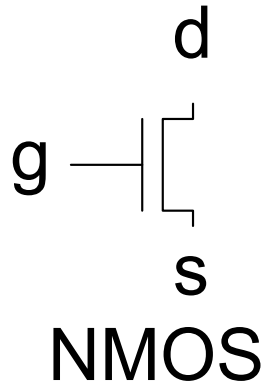
- ◆ If the voltage at the gate is removed, electrons are not attracted to the area between the source and drain. The pathway is broken and the transistor is turned off.

關



A Working Transistor (6/6)

- ◆ There are two major types of MOSes (switches), which operates in different conditions
 - NMOS: $g=1$, switch **on**; $g=0$ switch **off**
 - PMOS: $g=1$, switch **off**; $g=0$ switch **on**



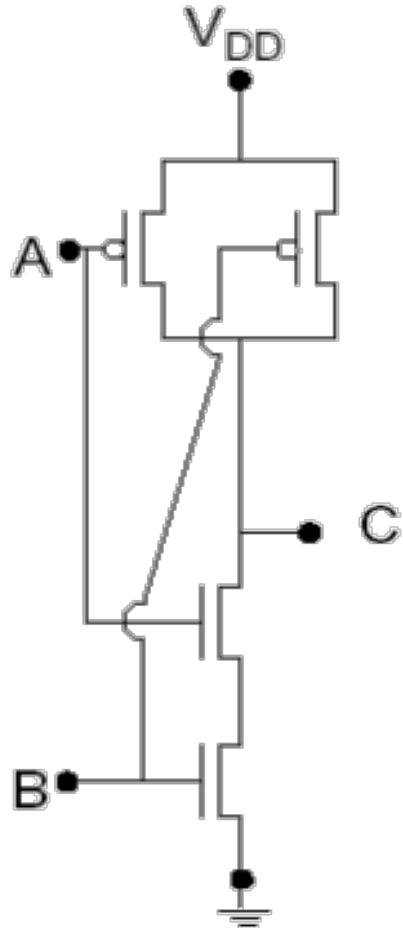


相關電壓電流特性
及電路分析等知識
我們是在_____課中介紹的

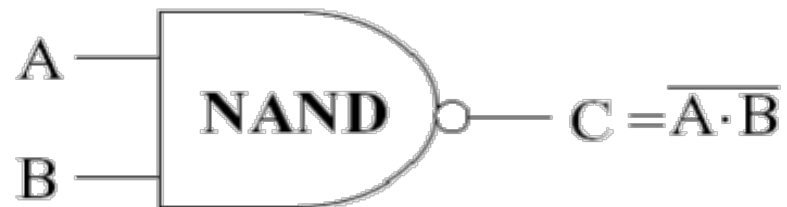
答：「數位電子導論」、「超大型積體電路設計」

有了開關就可以做邏輯閘

◆ CMOS NAND:

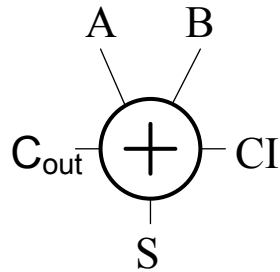


A	B	$A \cdot B$	$C = \overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

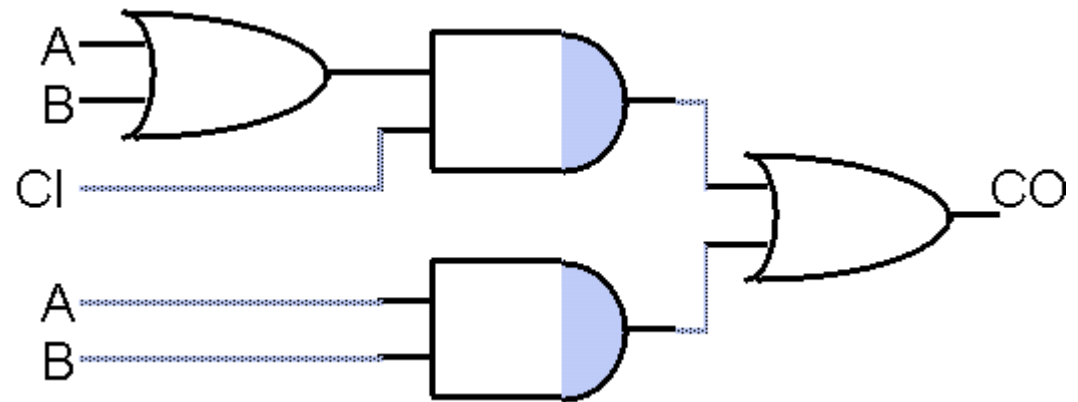
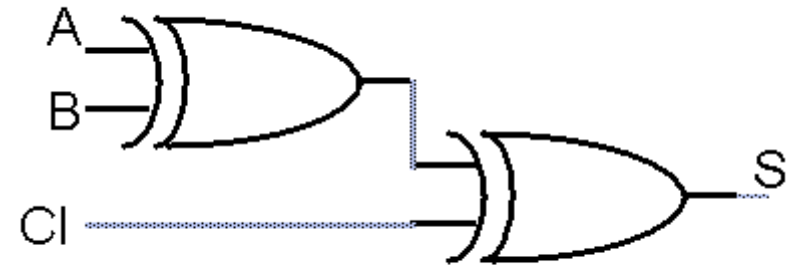


有了邏輯閘就可做邏輯電路

◆ 加法器：

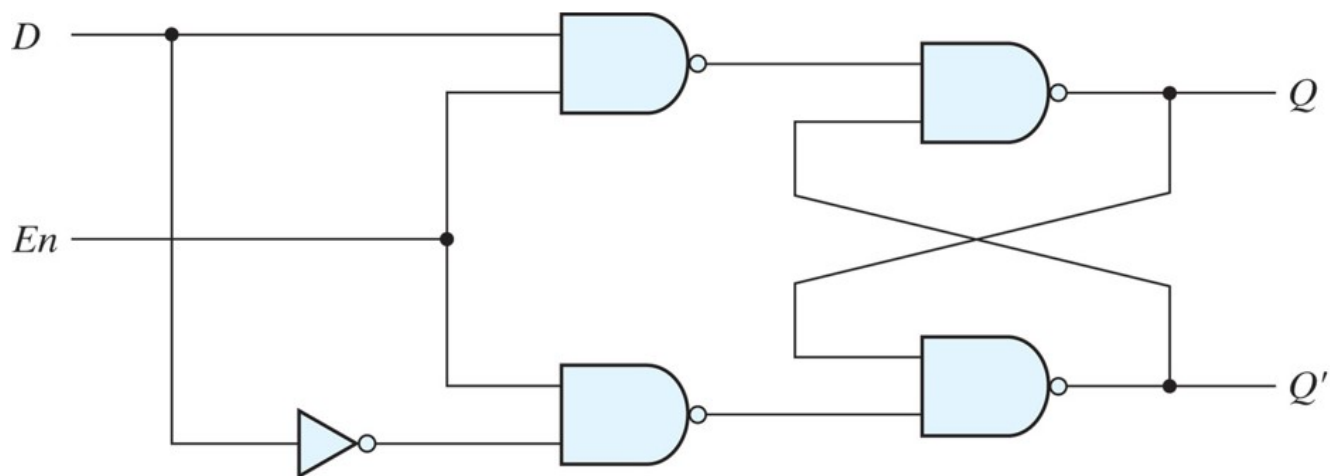
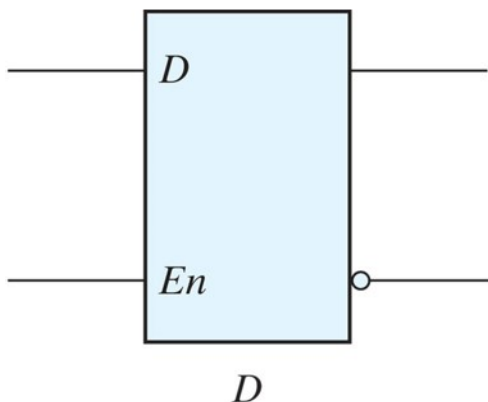


A	B	CI	C _{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



也可以做記憶元件

- ◆ 可以儲存一個bit的元件 (如: Latch, Flip-flop) :



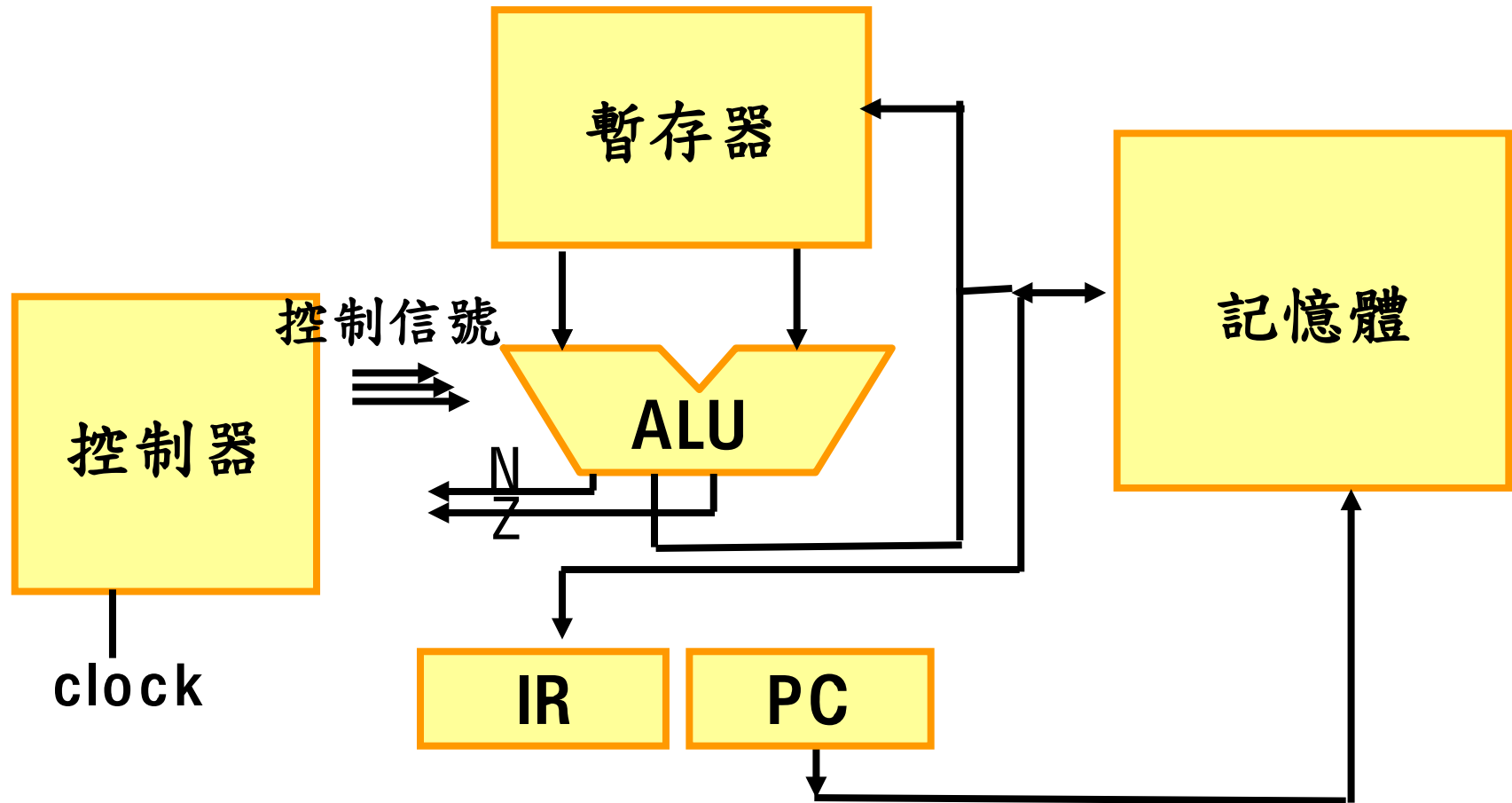
En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state



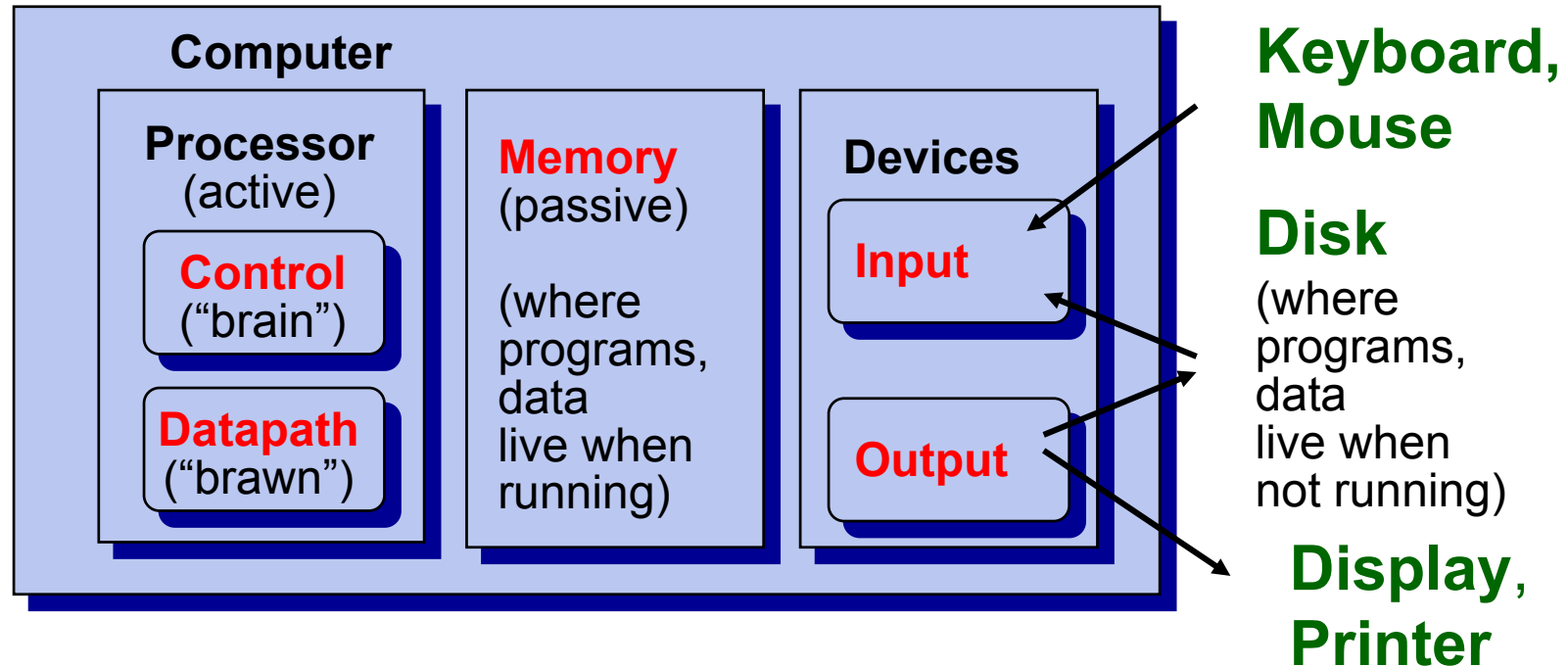
這部份的學問叫_____

答：「邏輯設計」、「數位系統設計」

於是電腦的主要部份都可以做了



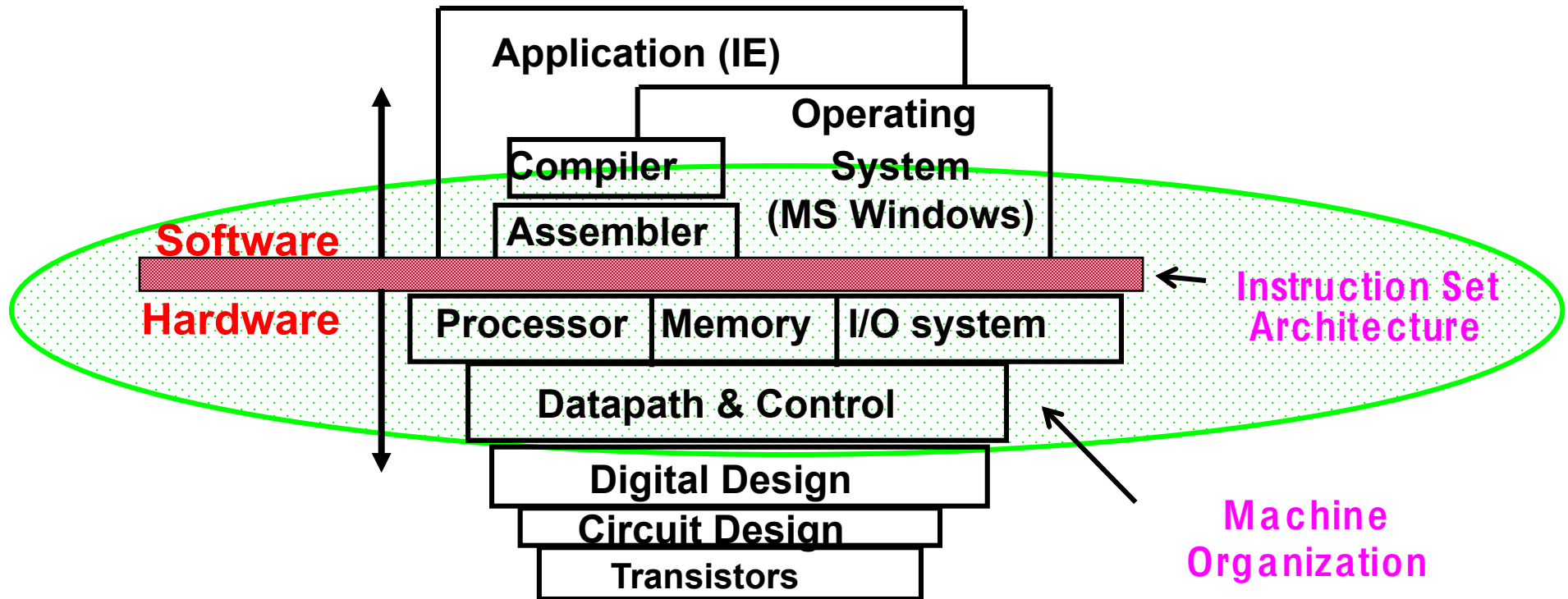
Basic Organization of Any Computer



Computer Organization

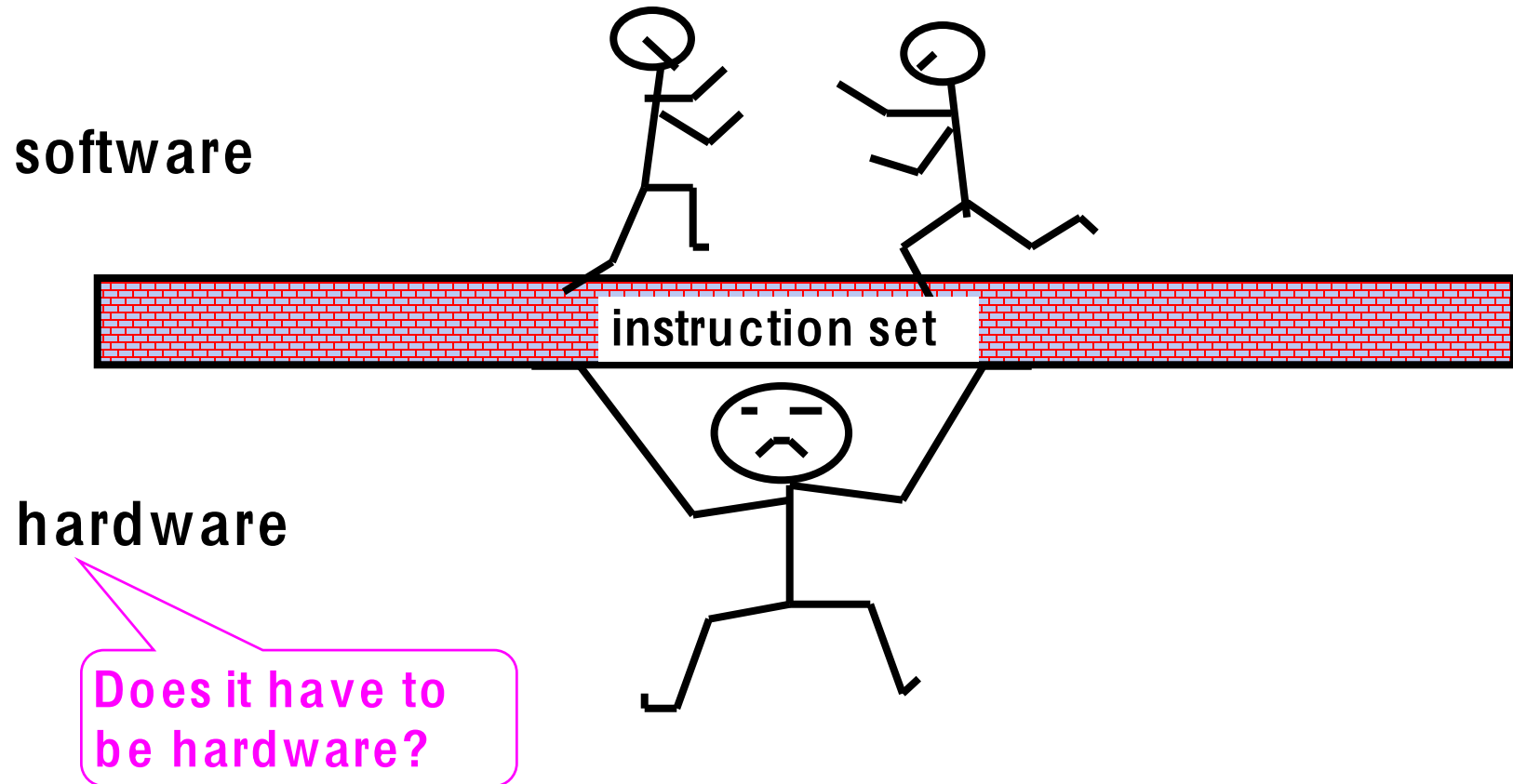
- ◆ Capabilities and performance characteristics of principal functional units, ex: registers, ALU, shifters, ...
- ◆ Ways in which these components are interconnected (*structure*)
- ◆ Information flows between components (*data, datapath*)
- ◆ Logic and means by which such information flow is controlled
- ◆ *Register Transfer Level* (RTL) description: A digital system is specified at the register transfer level (RTL) when it is specified by:
 - The set of registers
 - The operations performed on the stored data
 - The control that supervises the sequence of operations

What is Computer Architecture?



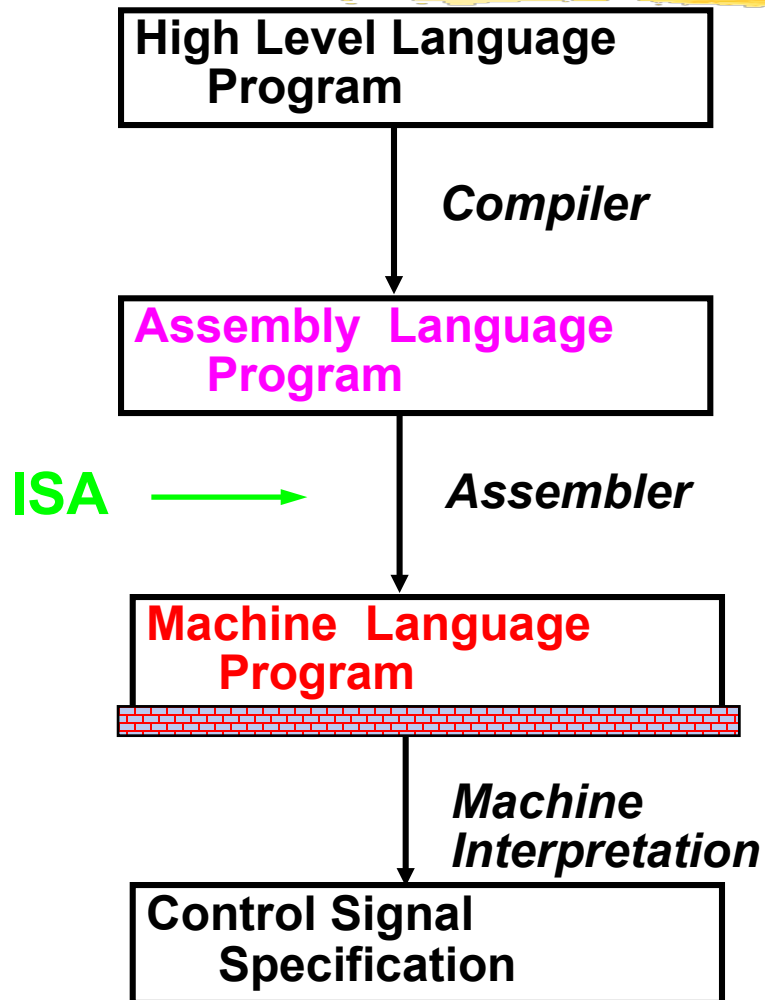
**Computer Architecture =
Instruction Set Architecture
+ Machine Organization**

Instruction Set as a Critical Interface



- ◆ Coordination of many **levels of abstraction**

Another Perspective



```
temp = v[0];  
v[0] = v[1];  
v[1] = temp;
```

```
lw      $15, 0($2)  
lw      $16, 4($2)  
sw      $16, 0($2)  
sw      $15, 4($2)
```

```
1000 1100 0100 1111 0000 0000 0000 0000  
1000 1100 0101 0000 0000 0000 0000 0100  
1010 1100 0101 0000 0000 0000 0000 0000  
1010 1100 0100 1111 0000 0000 0000 0100
```

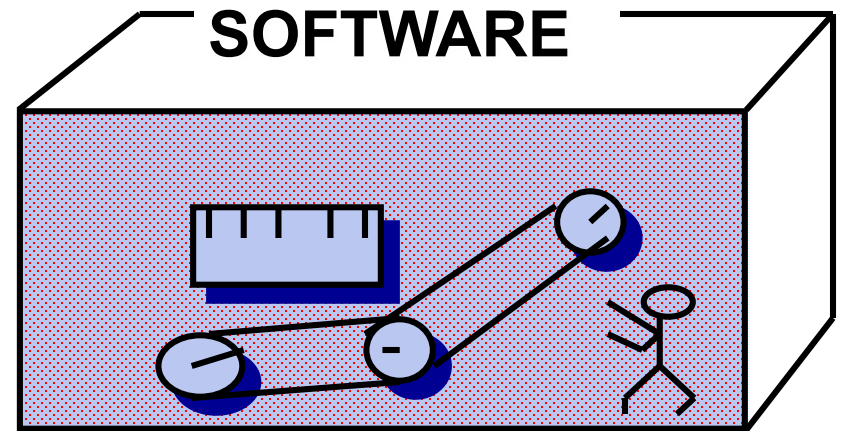
ALUOP[0:3] <= InstReg[9:11] & MASK

Instruction Set Architecture (ISA)

“... the attributes of a [computing] system as seen by the programmer, *i.e.* the **conceptual structure and functional behavior**, as distinct from the organization of the data flows and controls, the logic design, and the physical implementation.”

— Amdahl, Blaaw, and Brooks, 1964

- ☐ Organization of Programmable Storage
- ☐ Data Types and Data Structures: Encodings and Representations
- ☐ Instruction Set
- ☐ Instruction Formats
- ☐ Modes of Addressing and Accessing Data Items and Instructions
- ☐ Exceptional Conditions

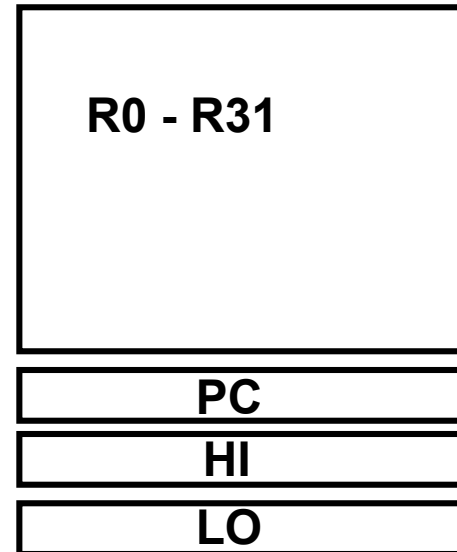


MIPS R3000 ISA

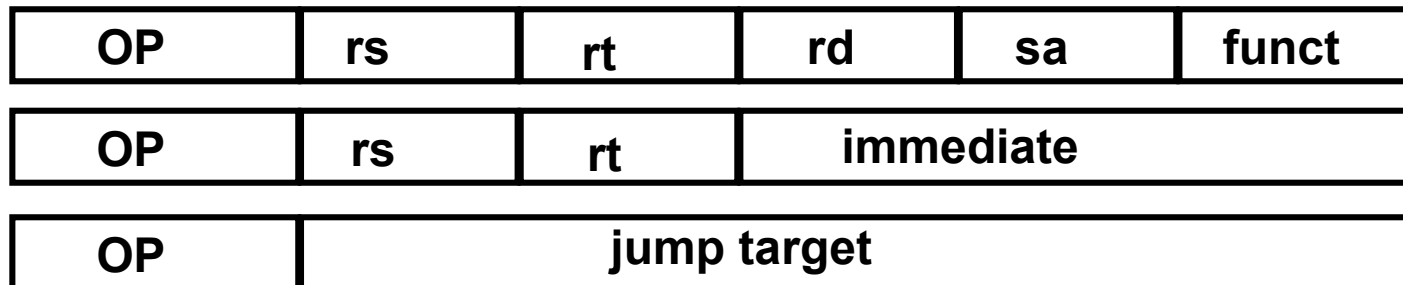
◆ Instruction categories:

- Load/Store
- Computational
- Jump and Branch
- Floating Point
 - coprocessor
- Memory Management
- Special

Registers



3 Instruction Formats: all 32 bits wide



ISA Examples

32-bit → 32/64-bit

HP PA-RISC	v1.1, v2.0	1986-96
Digital Alpha	v1, v3	1992-97
Oracle Sparc	v8, v9	1987-2019
IBM Power	POWER v1~v3.1	1990-2020
Intel x86	8086,80286,80386, 80486,Pentium, MMX, SIMD, IA-64, x86-64...	1978-2019
MIPS	MIPS I, II, III, IV, V, MIPS32/64 r6	1985-2014
ARM	ARM v1~v9.6-A	1985-2024
RISC-V	RISC-V v1~v2.2	2010-2021

Why Do Computer Architecture?

- ◆ **RAPID CHANGES**
- ◆ **It is exciting!**
- ◆ **It has never been more exciting! 、(*° ▽ ° *)ノ**
- ◆ **It impacts every other aspect of computer science and electrical engineering**

Course Administration

- ◆ 授課老師: 劉一字
 - 研究室: T4-305-2 電話: 3664
email: yyliu@mail.ntust.edu.tw

- ◆ 助教: 林廷奕
 - 實驗室: LB-002 電話: 6808
email: m11415042@mail.ntust.edu.tw

- ◆ CS2006:
 - 時間: 星期二 10:20-12:10, 星期三 9:10-10:00
 - 地點: 星期二 TR-313, 星期三 TR-409-2

- ◆ Office Hour:
 - 星期三 10:20-12:10
 - 考量大家可能有排課，請以e-mail提前預約時段 (可預約其他時段)

Prerequisite

- ◆ Prerequisite courses:
 - Digital System Design
 - C/C++ Programming

YouTube Playlists for your reference

<https://www.youtube.com/@YiYuLiu/playlists>

Text Book

Computer Organization and Design: The Hardware/Software Interface, 6th MIPS ed.,
David Patterson and John Hennessy, 2020



RISC, RAID

2017 Turing Award



Outline-28



**史丹福大學
前校長**

Computer Organization

Topics Covered

***Computer Organization and Design: The
Hardware/Software Interface, 6th MIPS ed.,
David Patterson and John Hennessy, 2020***

Topic	Chapter
Introduction	1
The Role of Performance	1
Instructions: Language of the Machine	2
Arithmetic for Computers	3
The Processor: Datapath and Control	4
Enhancing Performance with Pipelining	4
Exploiting Memory Hierarchy	5
Storage and Other I/O Topics	6
Multicores, Multiprocessors and Clusters	7

Expected Course Workload

- ◆ 4-5 project assignments
 - MIPS assembly, Booth's multiplication, Microarchitecture Simulation
- ◆ 1-2 Quizzes might be given in class
- ◆ One midterm and one final examination
- ◆ Grade breakdown in **points**, denoted as **%**
 - Project Assignments 50% (40 - 50%)
 - Quizzes 0% (0 - 10%)
 - Midterm Exam 25% (20 - 30%)
 - Final Exam 25% (20 - 30%)
 - Extra bonus points for Q&A in Moodle Discussion Forum
- ◆ Notice that the lump sum points could exceed 100%
 - In that case, final score is based on interpolation between 0 and 99

Code of Honor

- ◆ **What is cheating?**
 - It's simple: just be **HONEST!**
 - Study together in groups is encouraged
 - Work must be your own
 - Copy is definitely a cheating act
- ◆ **Cannot make examinations**
 - For fairness reason, it is unlikely to provide makeup examinations

Online Interactive Quiz

- ◆ 請帶手機、平板、筆電到教室
- ◆ 連線至App Store安裝Socrative Student Version
- ◆ <https://socrative.com/apps/>



- ◆ Room Name: NTUSTCA
- ◆ Your name: 學號 (英文要大寫)